

Experiment

‘p_4_2_d_adjacent_duplicates_seq_without_4th_ensures’

Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_2_d_adjacent_duplicates_seq_without_4th_ensures

Natural language statement: null

Method signature: p_4_2_d_adjacent_duplicates_seq_without_4th_ensures(inputs: seq<int>) returns (duplicates: seq<int>)

Ensures

- forall d :: d in duplicates ==> d in inputs
- forall i :: 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1] ==> inputs[i] in duplicates
- forall d :: d in duplicates ==> exists i :: 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1] && inputs[i] == d

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_2_d_adjacent_duplicates_seq_without_4th_ensures(inputs: seq<int>) returns (duplicates
↪ : seq<int>)`

The method should respect the following contract:

`ensures forall d :: d in duplicates ==> d in inputs, ensures forall i :: 0 <= i < |inputs|
↪ | - 1 && inputs[i] == inputs[i + 1] ==> inputs[i] in duplicates, ensures forall d
↪ :: d in duplicates ==> exists i :: 0 <= i < |inputs| - 1 && inputs[i] == inputs[i
↪ + 1] && inputs[i] == d`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnLeb1KjrN6zprYcpxhphnY8Iqgra

Model: gpt-5.5-2026-04-23

Created at: 1780653681

Finish reason: stop

Usage: 426 tokens in, and 6414 tokens out

```
{
  duplicates := [];
  var i := 0;

  while i < |inputs| - 1
    invariant 0 <= i <= |inputs|
    invariant forall d :: d in duplicates ==> d in inputs
    invariant forall d :: d in duplicates ==> exists j :: 0 <= j < i && j < |inputs| - 1
      ↪ && inputs[j] == inputs[j + 1] && inputs[j] == d
    invariant forall j :: 0 <= j < i && j < |inputs| - 1 && inputs[j] == inputs[j + 1] ==>
      ↪ inputs[j] in duplicates
    decreases |inputs| - i
  {
    if inputs[i] == inputs[i + 1] {
      duplicates := duplicates + [inputs[i]];
    }
    i := i + 1;
  }
```

```
}  
}
```

Final Program

```
method p_4_2_d_adjacent_duplicates_seq_without_4th_ensures(inputs: seq<int>) returns (  
  ↪ duplicates: seq<int>)  
  ensures forall d :: d in duplicates ==> d in inputs  
  ensures forall i :: 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1] ==> inputs  
    ↪ [i] in duplicates  
  ensures forall d :: d in duplicates ==> exists i :: 0 <= i < |inputs| - 1 &&  
    ↪ inputs[i] == inputs[i + 1] && inputs[i] == d  
{  
  duplicates := [];  
  var i := 0;  
  
  while i < |inputs| - 1  
    invariant 0 <= i <= |inputs|  
    invariant forall d :: d in duplicates ==> d in inputs  
    invariant forall d :: d in duplicates ==> exists j :: 0 <= j < i && j < |inputs| - 1  
      ↪ && inputs[j] == inputs[j + 1] && inputs[j] == d  
    invariant forall j :: 0 <= j < i && j < |inputs| - 1 && inputs[j] == inputs[j + 1] ==>  
      ↪ inputs[j] in duplicates  
    decreases |inputs| - i  
  {  
    if inputs[i] == inputs[i + 1] {  
      duplicates := duplicates + [inputs[i]];  
    }  
    i := i + 1;  
  }  
}
```

Total Token Usage

Input tokens: 426

Output tokens: 6414

Reasoning tokens: 6141

Sum of 'total tokens': 6840

Experiment Timings

Overall Experiment started at 1780653680997, ended at 1780653829121, lasting 148124ms (148.12 seconds)

Iteration #1 started at 1780653680998, ended at 1780653829120, lasting 148122ms (148.12 seconds)